

Supplementary Information — Paper E1 cod forecast ladder

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Forecast skill of structural models for Northern cod: a scored ladder

This supplement supports the main article (Abaee, *Does structural elaboration improve biomass forecasts?*, Northern cod forecast-ladder series). Section numbers are cited in the main text as SI-1, SI-2, and so on. All content here is derived from the analysis code and result files archived with this study; no result appears in this supplement that is not reproducible from that code.

SI-1 Order in which the passes were run

The main text (Section 4) states that the study is a fixed computational protocol rather than a prospective registration, that no dated pre-scoring protocol file exists, and that the additional passes were declared in the text as they were added. This section records the order of execution, documenting which choices were fixed before scoring and which were not.

The order below is established from the analysis code and its recorded metadata, not from recollection. Passes 1 and 2 are labelled as such in the driver script `run_ladder.py` and are distinguished in the archived `results/meta.json` by the keys `catch_pass1` and `catch_pass2`. Passes 3 and 4 are separate scripts that import from the pass-1/2 driver `run_ladder.py`, which fixes their order after it: `run_xte.py` imports the model definitions (SPECS) themselves, while the capelin scripts import the shared machinery (DATA, OUT, EPS, step, naive_baselines) and add one further model of their own, `M_cap_index`.

#	Pass	Catch / data treatment	Produced by	Archived output
1	Regime catch	Regime constants 240/120/5 kt, read from the SAR prose	run_ladder.py (# Pass 1 (regime C))	fixed_window_scores.csv, rolling_summary.csv (rows catch = regime)
2	Annual landings	Annual series, Schijns et al. (2021) Table 1, tonnes/1000	run_ladder.py (# Pass 2: annual catch)	same files, rows catch = annual
2b	Survey start	Forecast initiated at the survey state, annual catch	run_survey_start.py called within the pass-2 block of run_ladder.py	survey_start_forecasts.csv
3	Capelin	Continuous capelin-W ablation, observed acoustic years only	run_capelin_index.py, run_capelin_regime.py	capelin_index_summary.csv, capelin_regime_summary.csv
4	Specification B	The ladder on the xteNCAM domain alone, not pooled with NCAM 2016	run_xte.py	xte_rolling_summary.csv, xte_fixed_window_scores.csv

Three points follow from this order and bear on how the scores should be read.

The five ladder models were fixed before any pass was scored. The SPECS list in `run_ladder.py` defines M1, M1b, M2, M3 and M4 once, and Specification B (pass 4) imports that same list rather than redefining it. None of M1–M4 was altered after a score was seen. The capelin pass is the one place where a model was added after scoring began: it introduces `M_cap_index`, a prey-informed variant, which is reported in the main text as its own ablation and is not part of the five-rung ladder.

Two elements of Definition 2.4 were formalised after the Section 3 scores had been computed: the 5% tie band, and the comparator declarations for the non-nested rungs. This is the substantive qualification on the protocol, stated in the main text. Neither element changes any retention outcome: the tie band never decides a retention result, the smallest deficit being 17%.

Specification B was run last and was never pooled with the NCAM 2016 domain. The constraint is recorded in the code itself, whose module docstring reads “Wave E ladder on Ω_{xte} alone. Do not pool with NCAM 2016.”, and in `meta.json`, which records that the full `xteNCAM` SSB table was not extracted and was not pooled. The two specifications are therefore scored separately throughout, and no claim in the main text transfers a result from one to the other.

SI-2 Schaefer is not the (s) limit of the depensation family

Statement. In the family of Definition 2.1, the Schaefer law corresponds to the choice (a). The limit (s) in the depensation branch yields $(a(S_t) = S_t/K)$, hence a cubic modification of the logistic surplus, and not the Schaefer law.

Proof. Substituting $(s = 0)$ in $(a(S_t) = (S_t - s)/(K - s))$ gives $(a(S_t) = S_t/K)$, so $(g(S_t) = r S_t (1 - S_t/K)(S_t/K))$, a cubic surplus term. The Schaefer law requires (a), which is the separate branch in which the depensation factor is replaced, not obtained by a limiting value of (s). ■

The distinction matters because it fixes what M1b adds to M1: a separate branch of the family, not a limiting case of it. Abaee (2026b) makes the same distinction.

SI-3 Uncertainty layer: bandwidth grids, disagreement rows, and provenance

Bandwidth and block-length sensitivity

The M1-against-persistence margin was recomputed over a grid of HAC truncation lags and moving-block bootstrap block lengths, because the automatic $(h-1)$ truncation and the $((h,3))$ block length are not neutral choices when origins overlap at $(h=5)$, the estimation window expands, the predictand is a smoothed reconstruction, and the sample straddles a structural break.

Quantity	Specification A	Specification B
DM (z), $(h=1)$, lag 0 $\rightarrow$ lag 3	1.14 $\rightarrow$ 2.30	2.81 $\rightarrow$ 1.70
Bootstrap (p), $(h=1)$, block 3–9	0.17–0.19	0.036–0.049
Bootstrap (p), $(h=5)$, block 3–9	0.26–0.28	0.008–0.026

The DM statistic crosses the conventional 1.96 threshold within the range of defensible bandwidths on both specifications — upward on A, downward on B — so no verdict rests on it. The bootstrap is markedly more stable. The qualitative reading in the main text (Specification A within noise, Specification B separating) is therefore a bootstrap result rather than a DM result. The grid was run for the M1-against-persistence margin; robustness across block lengths is established for that comparison and is not asserted for every module.

The four disagreement rows

Four of the twenty-eight rows of Table 9 have a bootstrap interval excluding zero while $(|z| < 1.96)$:

Spec	(h)	Module	Comparator	DM (z)	95% CI (kt)
A	1	M4	M3	0.99	[+4.7, +144.7]
B	1	M3	persist	1.85	[+1.0, +92.5]
B	5	M1b	persist	1.80	[+18.2, +250.9]
B	5	M4	M3	1.88	[+20.2, +177.4]

Both summaries are reported side by side and no verdict rests on either where they disagree.

Provenance

The layer is produced by `campaign_e1_dm_uncertainty.py`, deterministic under seed 0, with output archived at `results/e1_dm_uncertainty.csv`.

SI-4 Log-floor binding counts

The log-RMSE scores of Table 4 use $((S, \{f\}))$ with $(\{f\} = 10^{\{-3\}})$ kt, the trajectory code clipping the state to $([_ \{f\}, 10^{\{6\}}])$ kt. The floor binds on the following counts of origins, taken from the archived per-origin records (Specification A: the annual-landings rolling pass of Section 3.2).

Spec	(h)	origins	M1	M1b	M2, M3, M4
A	1	25	15	17	M3 at 3; M2 and M4 at most once per horizon
A	5	21	19	19	M3 at 3; M2 and M4 at most once per horizon
B	1	59	22	24	between 0 and 11
B	5	55	36	46	between 0 and 11

The raw-RMSE column, not the log column, is the retention score, so these counts qualify a reported diagnostic rather than the verdict.

SI-5 Retention-rule operating characteristics: full decomposition

Design and results are archived in full as E1_SIMULATION_RESULTS.md, with the pre-registration at wave_e_cod/SPECIFICATION_v4.md (locked before execution) and the per-replicate output at wave_e_cod/results/sim_retention_power.csv (10,000 rows). This section records the detail condensed out of Section 3.7.

SI-5.1 Where the power is lost

H2 alone is “beats persistence by more than the 5% band at both horizons”, ignoring the comparator gate.

DGP	truth	passes H2 alone	passes full rule	gates, absolute	gates, conditional on H2
D1	M1	0.975	0.975	0.000	0% removed
D2	M1	0.420	0.420	0.000	0%
D3	M2	0.328	0.100	0.228	69%
D4	M1b	0.180	0.010	0.170	94%

The absolute cost of the comparator gates is therefore 0.17–0.23 of the shortfall on the two cells where they bind.

Both framings hold. In absolute terms the dominant failure is H2 – the true module often does not out-predict persistence on data it generated. Conditional on clearing H2, the comparator requirement is a dominant further filter.

SI-5.2 The chance baseline

With five structural modules, random assignment would place the generating module first 20% of the time.

DGP	true module has lowest h=1 RMSE	versus chance
D1 (M1 true)	0.627 (62.7%)	far better
D2 (M1 true)	0.645 (64.5%)	far better
D3 (M2 true)	0.0375	worse than chance
D4 (M1b true)	0.258 (25.8%)	barely better

On D3 the stock-flow module’s mean one-year error (85.7 kt) exceeds that of the residual (77.9 kt) and Allee (79.9 kt) modules. Estimation noise actively disadvantages the correct structure relative to its siblings at this sample size.

SI-5.3 Alternative decision rules

Computed on the same archived replicates, no refitting.

Decision rule	mean power	specificity
Retention rule as adopted (H1–H3, 5% band)	0.376	0.978
Without the comparator gate (H2, H3 only)	0.476	0.972
Persistence only, h=1, 5% band	0.562	0.955
Persistence only, any margin	0.542	0.765
Full rule, no tie band	0.436	0.772
Full rule, 10% band	0.337	0.998
MASE < 1 against the naive benchmark	0.651	0.675
Information criterion, $n \cdot \log \text{MSE} + 2k$	0.509	0.992

False-retention across the four structural DGPs is 0.044 per module-replicate pair, or 0.176 falsely retained modules per replicate; under the persistence-true null the per-module rate is 0.005.

SI-5.4 Two failed pre-check candidates

A diagnostic that told an analyst in advance whether the rule has power on a given series would be more valuable than any rule variant. Two were examined and neither works.

1. **Dispersion of scores across modules.** Median coefficient of variation against true-module power: Spearman $\rho = 0.52$. Too weak to license a protocol.
2. **Margin of the best-scoring module over persistence.** Spearman $\rho = 0.88$, Pearson $r = 0.94$ — strongly predictive, but it fails on two grounds. It is computed from the same out-of-sample scores the retention rule consumes, so it cannot gate the decision; and thresholding it at the tie band misclassifies both D3 cells, flagging “apply” where the generating module wins less often than chance.

Identifying when the rule has power therefore remains an open problem.

SI-6 Reproducibility package

Everything in the main text and in this supplement recomputes from the registered scripts and result files archived at <https://github.com/MIKEAA2020/general-sustainability>. The inventory below is the complete computational surface of the cod paper; checksums (md5, first 8 hex) pin the exact file states scored at release, and every entry is deterministic — re-execution in a fresh environment regenerates the archived outputs byte for byte.

A self-contained zip (E1_REPRODUCIBILITY_PACKAGE_v62_20260918.zip, distributed alongside this supplement) mirrors the directory layout so that every script below runs unmodified out-of-go-tree on numpy/pandas/scipy, includes the frozen reference outputs for byte-level diffing, and ships a `verify_package.py` that re-executes the survey-target campaign and the DM layer in a scratch directory and compares md5s against the archived files.

Script (repo path)	Role in the paper	md5	
wave_e_cod/src/ run_ladder.py	Passes 1 (regime catch) and 2 (annual landings); model ladder scoring; writes meta.json keys catch_pass1 / catch_pass2	db6b4ae6	results fixed_window_scores.csv, rolling_summary.csv, meta.json
wave_e_cod/src/ run_xte.py	Pass 3: xteNCAM extended-series scoring (imports SPECS from run_ladder.py)	5a5229fd	results xte_* scores
wave_e_cod/src/ run_capelin_index.py	Capelin-index intervention module (scored against origin-matched persistence)	fa648aab	capelin index scores
wave_e_cod/src/ run_capelin_regime.py	Capelin wind-era variant	4fda8885	capelin regime scores
arena_agent_1/ other documents/ rerun_campaigns/campaign_e1_base	Recomputed naive baselines, origin-matched, asserted against frozen values	0159acd0	verified baseline print-outs
arena_agent_1/ other documents/ rerun_campaigns/campaign_e1_m1_recon	M1 fit reconciliation audit	a97aed57	reconciliation output
batch 7 (audits of agent arena 1 paper rewrites)/campaign_e1_dm_uncertainty.py	Post-freeze DM + moving-block bootstrap layer (Spec A/B), seed 0, 20,000 replications, deterministic	81c559b8	results/ e1_dm_uncertainty.csv
batch 7 (audits of agent arena 1 paper rewrites)/campaign_e1_dm_uncertainty	Regime-stratified DM layer	6ced3555	results/e1_dm_uncertainty_regime.csv

Execution order. run_ladder.py (Pass 1 then Pass 2) → run_xte.py (Pass 3) → run_capelin_index.py / run_capelin_regime.py (Pass 4) → campaign_e1_dm_uncertainty.py and campaign_e1_dm_uncertainty_regime.py (post-freeze audit layers, read-only on scores). campaign_e1_baselines.py and campaign_e1_m1_reconciliation.py are audit reruns: they reassert the frozen values, they never re-decide them.

Dependency claim. No script requires external credentials; the only non-redistributable raw input is the DFO assessment table itself, whose registered transcription is archived as wave_e_cod/data/ncam_2016_table_a2.csv and wave_e_cod/data/xtencam_table17_ssb.csv.

SI-7 Autumn survey index as forecast target (companion to §3.8)

Object. The DFO autumn research-vessel trawl survey abundance index for 2J3KL cod, transcribed from Table 3 of Schijns et al. (2021), which sources the series from the Autumn DFO research-vessel survey results table: 33 consecutive annual values 1983–2015, no missing years (verified by a continuity assertion inside the campaign script). The index is converted to kt-equivalents by the deterministic, declared unit-reconciliation constant ($q^{[t]} = \{_{1983:2015}(_t/_1) = 5.702\{-4\}$) kt per index unit, computed once from the full overlapping sample and never refitted per forecast origin; the resulting target spans 12.4–1,427.0 kt-equivalents. **Correction to v4 (flagged sensitivity):** margin magnitudes are *not* invariant to $(q^{\{ \# \}})$ — the optimizer (K)-bounds $([_ \{ \} + 10, 5000])$ kt track the rescaled

state, so fitted paths shift slightly. What is invariant is every verdict: re-scoring the whole ladder (annual landings treatment) at the interquartile bounds of ($_t/I_t$) ($q = p25/p50/p75$) leaves all H1/H2/H3 verdicts and the empty retained set identical. Sensitivity table (closest-structural margin vs persistence, % of RMSE):

q level	h = 1: closest deficit	h = 5: closest deficit	Any model retained?
p25	+22.6%	+27.8%	No
p50 (= 5.702×10^{-4})	+29.8%	+42.7%	No
p75	+32.8%	+46.8%	No

Script `campaign_e1_survey_qsensitivity.py`; output archived as `results/e1_survey_qsensitivity.csv` in the reproducibility package.

Pre-registration. The design was frozen before any score was read (`SPEC_E1_SURVEY_TARGET_20260918.md`): object, origin sets, uncertainty-layer conventions (DM with Newey–West HAC lag $h-1$, moving-block bootstrap block $\max(h,3)$, 20,000 replications, seed 0), declared departures (no Brier score — the index carries no normative threshold; no survey-start variant; no capelin modules, which are not part of the frozen ladder), and a pre-registered hypothesis (smoother-injected autocorrelation favours persistence, predicting a shrunken or inverted persistence advantage on a noisier target).

Machinery reuse and fidelity proof. The campaign imports `run_ladder.py` unmodified and executes its rolling engine on the survey target for both catch treatments. Before any survey number is written, the engine is re-run on the Spec A series and must reproduce the frozen values (naive persistence 98.05 kt at $h = 1$; stock-flow regime-catch 144.09 kt at $h = 1$); assertions enforce this. Two consecutive campaign executions produce byte-identical output files. Computed origin lists equal [1990..2014] ($h = 1$, $n = 25$) and [1990..2010] ($h = 5$, $n = 21$) — origin-for-origin identical to Spec A.

Artifacts (archived in the reproducibility package). `results/e1_survey_target_rolling_forecasts.csv` (per-origin folds, both treatments), `results/e1_survey_target_rolling_summary.csv`, `results/e1_survey_target_fixed_window_scores.csv`, `results/e1_survey_dm_uncertainty.csv` (gap, DM z , bootstrap CI and p), plus the campaign script `campaign_e1_survey_target.py`.

Full survey-target table (annual-catch treatment; kt-equivalents).

Model	h = 1 RMSE	h = 5 RMSE	h = 1 MAE	h = 5 MAE
naive persist	120.5	249.2	52.9	118.6
naive train mean	482.4	553.0	429.9	490.2
M1_autonomous_Schaefer	184.2	386.6	90.4	188.2
M1b_autonomous_Allee	156.3	355.5	92.2	176.9
M2_stockflow_regimeC	224.9	671.4	98.4	562.8
M3_AR_residual	290.3	716.7	129.5	608.6
M4_delayed_info	371.3	880.5	203.0	803.1

Under the coarse-regime catch treatment the ladder spans 156.7–357.1 kt-equivalents at $h = 1$ and 355.5–913.5 at $h = 5$ against the same origin-matched persistence baselines (120.5/249.2). Uncertainty layer: every structural deficit vs persistence separates from zero at both horizons except the stock-flow module at $h = 1$ (annual treatment gap +36,090 kt², 95% CI [−78, +80,933], $p = 0.0615$; the sign is a deficit either way). Fixed windows: collapse (train 1983–1990, score 1991–1995) structural RMSE 1,102–1,274 vs 791.6 persist / 863.8 mean; recovery (train 1995–2007, score 2008–2015) structural RMSE 41.9–120.1 annual / 33.1–93.2 regime (the upper end is the Allee module, which misses persistence’s 109.4 under the annual treatment but beats it under the coarse regime), vs 109.4 persist / 243.1 mean — both windows mirror the reconstruction’s ordering (§3.8).

Information-set declaration. The survey value for year (t) is assigned as available at origin (t). The index is compiled after season close; the residual timing conservatism (an analyst scoring in January of year ($t+1$)) may

not yet hold the released table) is declared in the frozen design and is immaterial to the verdicts, which lose to persistence by large margins, not by timing shifts.

SI-8 Extended interpretive passages

This appendix carries, verbatim, two pairs of extended interpretive passages that were shortened in the main text for length: the fixed-window narratives (which certify the readings of Table 3 and of the archived Specification B window scores) and the depensation-boundary and scope-limit observations (which certify how M1b’s algebra behaves at data boundaries and which claims the autonomous-map proposition does not make). Material considered for this appendix but not reproduced here: the original Diebold–Mariano “Readings” narrative (its content is subsumed by the seeded 20,000-replication uncertainty layer of SI-3), the implied-maxima and net-production analyses (present in full in the main text), the parameter-estimation and log-RMSE floor narrations (carried by Appendix A.2/A.4 and SI-4), and the Edwards Aquifer cross-system paragraph (a single cross-reference sentence suffices at this venue). All numerals in the restored text reproduce exactly from the manuscript’s frozen data and result files or from direct recomputation with the registered engine (e.g., the terminal disclosure’s two one-step least-squares minima, 128.35 vs 127.84 kt², reproduce exactly from `run_ladder.py` at the stated constant catches 5.0 and 3.19 kt).

8.1 Fixed-window narratives, Specifications A and B

On Specification B the collapse window is train 1954–1989, test 1990–1995, and the recovery-stall window is train 1995–2012, test 2013–2024. As on Specification A, 1995 appears both in the collapse test window and at the start of the later training window; the two windows are scored independently and no fitted quantity crosses between them, so the shared year is not leakage. Catch is Regular et al. (2025) Table 1 landings (2024 persisted from 2023). No row is taken from NCAM 2016. The ladder’s rolling origin on Specification B uses a minimum of twelve training years ($n = 59$ at $h = 1$, $n = 55$ at $h = 5$); the naive baselines retain the eight-year minimum ($n = 63$ and $n = 59$). The two origin sets therefore overlap but are not identical, so a baseline computed over the longer naive origin sets would mix four extra early origins into the comparison. Table 6 accordingly reports the origin-matched baselines, computed on the ladder’s own origins: on the identical twelve-year origins persistence reads 84 kt ($n = 59$) at $h = 1$ and 300 kt ($n = 55$) at $h = 5$, and the training mean 458 kt and 522 kt, against M1’s 120 kt and 432 kt. Persistence remains the lowest-RMSE forecast. The mixed-origin reading (88 kt, quoted in the Table 6 caption) exceeds the controlled reading of 84.4 kt by 3.6 kt — an origin-mix effect of no retention consequence. On the identical post-break origins of the Table 7 control ($n = 33$, $h = 1$) the recomputed baseline is 75 kt, matching the post-break value reported with the two-regime control of Section 3.4. On the index modules’ origin sets (Table 8), persistence reads 97 kt ($n = 24$, $h = 1$) and 193 kt ($n = 20$, $h = 5$) on Specification A, and 79 kt ($n = 36$) and 288 kt ($n = 32$) on Specification B. The index module loses to the origin-matched baseline in every cell, and the verdict — the module is not retained — is unchanged.

8.2 Window numerical disclosure (annual vs. coarse-constant catches)

One numerical disclosure travels with the window: the annual-landings M1 (264 kt) and M1b (78 kt) differ from the coarse-regime recovery rows (M1 = 120, M1b = 90) although both treatments sit on the same SSB column. The reconciliation is that M1’s constant catch is not estimated on that column. It is the training mean of the catch file: 5.0 kt for the coarse regime, and 3.19 kt (the 1995–2007 mean of the annual landings) for the annual treatment. The two least-squares objectives are nearly flat at their minima (MSE 128.35 versus 127.84 kt²), so the (r, K) minimizer slides along the valley as the constant changes: $r = 0.458$ with K resting at the multi-start initialiser of 500.0 kt — not a bound, the lower bound on this window being 50.8 kt — against $r = 0.370$ with K at its upper bound, 5000.0 kt. The optimiser leaves K at its starting value in this fit; the profile of Section 3.2, not the optimiser’s behaviour, is what measures the flatness. The same geometry explains the M1b pair (r pinned at its upper bound in both treatments; $K = 105.8$ against 129.8 kt): the dependence of the M1b fit on the catch file is the flat-objective identification of a two-parameter autonomous fit on a 13-year window, not an inconsistency between fitting objects. The same flat valley caps what the ranking can resolve. Refitting the recovery-window Schaefer map with K fixed anywhere on [60, 5000] kt and r re-optimised moves the one-step training objective only from mean squared error 127.4 to 149.9 kt² (training RMSE 11.29–12.24 kt, a spread

under 0.95 kt), while r compensates from 0.435 to 0.773. On this window the (r, K) pair is not identified, and the ordering of valley variants is not a robust ranking.

8.3 Depensation boundary identifiability

Observation 3.2 (Boundary Allee estimate on the recovery window). Any $\mathfrak{s}$ strictly above the training minimum makes the depensation factor $a(S_t) = (S_t - \mathfrak{s})/(K - \mathfrak{s})$ negative for observations below $\mathfrak{s}$, so wherever the observed ΔS is positive at such a state the fit incurs a large squared residual. On the 1995–2007 recovery window seven of the twelve annual increments are positive — the five negative ones fall in a single consecutive run, 1999–2004 — and on balance this penalty pushes the estimate downward: the fitted $\mathfrak{s}$ descends towards zero (2.1×10^{-23} under the coarse regime and 9.4×10^{-6} under annual landings) while r reaches its upper bound of 2 and K settles just above the training range at 105.8 kt. Zero is excluded by the objective’s feasibility restriction $0 < \mathfrak{s} < 0.8K$, so these values are a numerical approach to an unattained infimum rather than an attained boundary estimate. At such values the fitted object is effectively the zero-threshold cubic branch $a(S) = S/K$ of Definition 2.1, not a positive-threshold Allee model, so M1b’s lower error here is not evidence for depensation: r and K absorb what $\mathfrak{s}$ does not.

8.4 Scope limits of the autonomous-map proposition

Two limits on the scope of this proposition should be stated explicitly, because both are easy to overstate. First, it does not say that autonomous maps cannot produce non-monotone paths: the fitted map has $F'(S^*) \approx -0.39$ at its attractor, so approach to the attractor is by damped oscillation, and declines followed by recoveries are the generic local behaviour there (from 950 kt the fitted map gives $950 \rightarrow 857 \rightarrow 899$ kt). What is excluded is recovery after a fall below the lower repeller, which is the feature the observed collapse path actually has. Second, the result is a statement about autonomous noise-free scalar maps only. It does not apply to the stochastic map of Definition 2.1 with $\varepsilon_t \neq 0$, nor to M3 and M4, which carry a residual state and a stale initial state respectively and are therefore not autonomous scalar maps. The proposition applies directly to M1 under the deterministic forecast convention used here, and to M1b whenever its own fitted map has the stated equilibrium structure. It does not formally cover M2, whose prescribed year-by-year catch makes the update a time-indexed family F_t rather than a single autonomous map; for M2 the argument is motivating rather than binding, and the collapse-window result of Section 3.1 is the relevant evidence. It is not an impossibility theorem for surplus-production modelling in general.